Observer as Constraint:

A Field-Theoretic Placement Across Quantum and Relativistic Regimes

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Abstract

The observer occupies an unresolved position in both quantum mechanics and general relativity. In quantum mechanics, definite outcomes are registered relative to measurement interactions, but the observer is not internally defined within the formalism. In general relativity, physical description is frame-relative, while the observer is typically represented only in idealised kinematic terms. This paper proposes a field-theoretic placement of the observer within the VFD event-order framework.

An observer is defined as a bounded, dynamically stable, self-referential substructure within the event set, characterised by a constraint functional $\mathcal{C}_{\mathcal{O}}$ that enforces internal coherence. The observer constrains the set of admissible field configurations: measurement corresponds to constraint-induced stabilisation, not external collapse. Relativistic frame dependence corresponds to the observer's embedding within the event partial order. Both roles — quantum state selection and relativistic frame specification — are unified as aspects of a single constraint-placement mechanism.

The paper connects this placement to the crystallisation dynamics (selection as constraint satisfaction) and to the GR programme (observer as linear extension with sampling measure). A brief section addresses implications for the locus of conscious experience within the framework, without claiming a full theory of consciousness.

This is a placement paper: it defines where the observer sits, what it does, and how it connects to both quantum and gravitational regimes. It does not derive a complete measurement theory, a consciousness theory, or new experimental predictions.

1 Introduction

The observer problem appears in three guises:

1. **Quantum mechanics.** In standard operational quantum mechanics, definite measurement outcomes are registered relative to measurement interactions, but the

formalism does not itself provide a universally agreed internal account of what an observer is or how it differs from any other physical system. The measurement problem is, at root, a placement problem: the observer is invoked but not located.

2. **General relativity.** Physical quantities are defined relative to an observer's frame, but the observer is modelled as an idealised worldline with no internal structure. Frame dependence is central to the theory, yet the entity that defines the frame is structurally empty.

3. **Consciousness.** The explanatory gap between physical dynamics and subjective experience is often framed as a separate problem. But it shares the same structural feature: the observer is assumed to exist but not placed within the dynamics.

Central proposal

> The observer is not external to the system and not an undefined primitive. It is a stable, self-referential, coherence-preserving substructure within the event-order geometry that imposes constraints on admissible configurations.

Scope

This paper defines the observer as a constraint structure, shows how this placement recovers quantum measurement and relativistic frame dependence, and connects to the crystallisation dynamics. A brief section addresses the locus of conscious experience. The paper does not derive a complete measurement theory, a theory of consciousness, or new experimental predictions. It is a placement paper: it identifies where the observer sits and what structural role it plays.

2 Limits of Formal Closure

Before defining the observer, it is necessary to address a common misapplication of Gödel's incompleteness theorems [Gödel, 1931] to physical theory.

2.1 What Gödel's theorems establish

Gödel's first incompleteness theorem states that any consistent formal system capable of expressing basic arithmetic contains statements that are true but not provable within the system. The second theorem states that such a system cannot prove its own consistency.

2.2 What they do not establish

The theorems apply to *formal symbolic systems* — axiomatic frameworks with specified rules of inference. They do not apply to physical dynamics, continuous systems, or geometric structures unless those systems are first encoded as formal axiomatic theories.

In particular:

- Gödel's theorems do not imply that physical reality is unknowable.
- They do not imply that consciousness is needed to "complete" a formal system.

- They do not establish limits on what physical systems can do — only on what formal proofs within a fixed axiomatic system can establish about themselves.

2.3 Relevance to observer placement

The relevant consequence is narrower but genuine:

> Gödel's theorems do not justify identifying the limits of a formal description with limits of the physical system described.

This motivates treating the observer as a *dynamical substructure* within the field, rather than leaving it as an undefined primitive in a formal encoding. The present argument does not treat the framework as a closed formal proof system of the Gödel type. The point is therefore not to evade incompleteness, but to avoid misapplying incompleteness results as if they directly constrained observer dynamics in physical theory.

3 Observer Definition

3.1 Prior definition

Paper XXIV [Smart, 2026c] defined an observer on the event poset $(\mathcal{E}, \prec)$ as a pair $O = (\leq_O, \mu_O)$: a linear extension (total ordering compatible with the partial order) together with a sampling measure (positive function assigning duration weights to events). This definition specifies what an observer *does* (orders and weights events) but not what an observer *is* as a physical substructure.

The present paper extends that definition.

3.2 Structural definition

Definition 1 (Observer region)**.** *An observer region is a connected subgraph $\mathcal{O} \subset \mathcal{G}$ of the transition graph, together with a constraint functional $\mathcal{C}_\mathcal{O} : \mathbb{R}^\mathcal{E} \to \mathbb{R}_{\geq 0}$ that measures the internal coherence of field configurations restricted to $\mathcal{O}$.*

Definition 2 (Observer)**.** *An observer is an observer region $(\mathcal{O}, \mathcal{C}_\mathcal{O})$ satisfying:*

1. ***Boundedness.*** *$\mathcal{O}$ is a proper subgraph of $\mathcal{G}$: $\mathcal{O} \subsetneq \mathcal{G}$.*

2. ***Stability.*** *The constraint functional is at a local minimum:*

$$\delta\mathcal{C}_\mathcal{O}[\Phi]\big|_{\Phi=\Phi_0} = 0, \qquad \delta^2\mathcal{C}_\mathcal{O}[\Phi]\big|_{\Phi=\Phi_0} > 0 \tag{1}$$

   for the ambient field configuration Φ_0. The first condition is stationarity; the second ensures genuine stability (the observer region lies in a basin of attraction, not at a saddle point).

3. ***Self-reference with boundary coupling.*** *The constraint functional $\mathcal{C}_\mathcal{O}$ depends primarily on the restriction of the field to $\mathcal{O}$, with secondary dependence on the boundary:*

$$\mathcal{C}_\mathcal{O}[\Phi] = \mathcal{C}_\mathcal{O}\big(\Phi|_\mathcal{O},\ \partial_\mathcal{O}\Phi\big) \tag{2}$$

   where $\partial_\mathcal{O}\Phi$ denotes the field values at the boundary of $\mathcal{O}$ (the neighbouring events in $\mathcal{G}$). The observer's coherence is determined primarily by its own internal state, with environmental coupling entering through the boundary.

4. ***Persistence.*** *The observer region $\mathcal{O}$ is stable under the event-order dynamics: it persists as a connected, coherent substructure across successive events in the partial order.*

Remark 3 (Relation to Paper XXIV)**.** *The Paper XXIV observer $(\leq_O, \mu_O)$ specifies the observer's epistemic access: what ordering and weighting it assigns to events. The present definition specifies the observer's physical substrate: what it is as a substructure within the event-order geometry. The two are compatible: an observer region $\mathcal{O}$ that satisfies Definition 2 naturally induces a linear extension (the ordering of events as encountered by $\mathcal{O}$) and a sampling measure (the duration weight determined by the local constraint structure).*

3.3 Functional properties

An observer as defined above has the following structural properties:

- **Coherence preservation.** The stability condition (1) ensures that the observer maintains internal consistency. Small perturbations do not destroy the observer structure.
- **Internal state coupling.** The self-reference condition (2) means the observer's coherence depends primarily on its own configuration. This is the structural basis for locality of experience.
- **Sensitivity to external configuration.** The boundary term $\partial_{\mathcal{O}}\Phi$ in (2) means the observer is sensitive to its environment through the boundary conditions imposed by the ambient field, without being globally determined.
- **Persistence.** The observer is not an instantaneous structure. It persists across the event ordering, maintaining coherence through successive events.

4 Observer as Constraint Mechanism

The central structural role of the observer is to constrain the set of admissible field configurations.

Definition 4 (Observer-admissible configurations)**.** *Given an observer $(\mathcal{O}, \mathcal{C}_{\mathcal{O}})$, the set of observer-admissible configurations is:*

$$\Phi_{\mathrm{adm}}(\mathcal{O}) = \left\{ \Phi \in \mathbb{R}^{\mathcal{E}} \;\middle|\; \mathcal{C}_{\mathcal{O}}\big(\Phi|_{\mathcal{O}},\, \partial_{\mathcal{O}}\Phi\big) \leq \epsilon \right\} \tag{3}$$

where $\epsilon > 0$ is a coherence threshold.

The observer does not select a configuration arbitrarily. It *constrains* the space of configurations to those compatible with its own internal coherence. Configurations that would destabilise the observer ($\mathcal{C}_{\mathcal{O}} > \epsilon$) are excluded from the admissible set.

Remark 5 (Not external selection)**.** *The observer is not an agent that reaches into the system and picks an outcome. It is a substructure whose existence imposes a boundary condition: only configurations compatible with the observer's coherence are dynamically accessible. This is constraint, not choice.*

Remark 6 (Constraint restricts multiplicity)**.** *If the unconstrained configuration space has N locally stable configurations (local minima of $\mathcal{F}$), the observer-admissible set $\Phi_{\mathrm{adm}}(\mathcal{O})$ will generically contain fewer than N configurations, since the additional constraint $\mathcal{C}_{\mathcal{O}} \leq \epsilon$ need not be satisfied at all minima of $\mathcal{F}$. The observer reduces multiplicity by excluding configurations incompatible with its coherence. A precise statement requires transversality or non-degeneracy assumptions on the joint level sets of $\mathcal{F}$ and $\mathcal{C}_{\mathcal{O}}$.*

5 Recovery of Quantum Measurement

The quantum measurement problem asks: how does a system in superposition resolve to a definite outcome upon measurement? Within the present framework, the answer is structural.

5.1 Superposition as configuration multiplicity

In the VFD framework, superposition corresponds to the coexistence of multiple admissible configurations (Paper XXVIII [Smart, 2026d], weak-ordering regime). The system has not yet resolved to a single stable configuration.

5.2 Measurement as constraint-induced stabilisation

When an observer $(\mathcal{O}, \mathcal{C}_{\mathcal{O}})$ interacts with a system in superposition, the observer's coherence constraint restricts the admissible configurations (Definition 4). The combined system — observed system plus observer — must satisfy:

$$\mathcal{F}[\Phi] \text{ locally minimal} \quad \text{and} \quad \mathcal{C}_{\mathcal{O}}\big(\Phi|_{\mathcal{O}},\, \partial_{\mathcal{O}}\Phi\big) \leq \epsilon \tag{4}$$

If only one of the original configurations satisfies both conditions, the system resolves to that configuration. If multiple configurations remain admissible, further constraint resolution (crystallisation) determines the outcome.

5.3 Structural mapping

Table 1: Quantum measurement mapped to observer-constraint resolution.

Standard QM	VFD observer-constraint
Superposition	Multiple admissible configurations
Measurement apparatus	Observer region $\mathcal{O}$ with constraint $\mathcal{C}_{\mathcal{O}}$
Collapse / state selection	Constraint-induced reduction of admissible set
Definite outcome	Stable configuration satisfying $\mathcal{F}$ and $\mathcal{C}_{\mathcal{O}}$
Born rule probabilities	Not derived (open)

Remark 7 (What is and is not recovered)**.** *The framework provides a structural account of how multiplicity reduces to a definite outcome: observer coherence constrains the admissible set. It does not derive the Born rule or the specific probability distribution over outcomes. Whether the Born rule emerges from the constraint geometry (e.g. through the stationary distribution of Paper XIV [Smart, 2026a]) is an open question.*

6 Recovery of Relativistic Frames

In general relativity, physical quantities are defined relative to an observer's frame. The VFD framework recovers this structure through observer placement.

6.1 Frame as observer embedding

Paper XXIV [Smart, 2026c] showed that different observers (different linear extensions of the event partial order) assign different time functions to the same event set. Simultaneity is observer-dependent, and elapsed-time assignments differ between observers.

In the present framework, this structure extends naturally:

- An observer region $\mathcal{O} \subset \mathcal{G}$ occupies a specific location within the transition graph.
- The observer's linear extension $\leq_O$ is determined by the order in which events become accessible from $\mathcal{O}$.
- The sampling measure μ_O is determined by the local constraint structure within $\mathcal{O}$.

A relativistic frame is not an abstract coordinate choice. It is the concrete embedding of an observer substructure within the event-order geometry.

6.2 No global frame

Paper XXIII [Smart, 2026b] proved that no canonical global clock exists on the event poset. In the observer-constraint framework, this has a structural explanation: no single observer region can span the entire event set while maintaining internal coherence (boundedness condition, Definition 2). A global frame would require a global observer, but the coherence constraint is inherently local.

6.3 Structural mapping

Table 2: Relativistic frame dependence mapped to observer embedding.

General relativity	VFD observer-constraint
Observer worldline	Observer region $\mathcal{O}$ persisting through event order
Reference frame	Linear extension $\leq_O$ induced by $\mathcal{O}$
Proper time	Cumulative sampling measure $t_O(e) = \sum_{e' \leq_O e} \mu_O(e')$
No global frame	No global observer (boundedness condition)
Frame-dependent quantities	$\mathcal{O}$-dependent admissible configurations

Remark 8 (Frames and consciousness)**.** *Every conscious observer defines a frame in this sense, but not every frame-defining substructure need be conscious. The structural requirements for frame definition (boundedness, persistence, induced linear extension) are necessary but not sufficient conditions for the kind of self-referential coherence discussed in Section 8.*

7 Integration with Crystallisation

The crystallisation operator $C[\Psi] = \text{argmin}_\Phi \mathcal{F}[\Phi]$ drives the system toward the stable attractor of the closure functional. The observer-constraint framework integrates with crystallisation as follows.

7.1 Observer as local constraint node

Crystallisation is a global process: the entire field configuration converges toward constraint satisfaction. The observer is a *local* constraint node: a substructure whose coherence requirement imposes an additional condition on the global resolution.

- **Crystallisation without observer**: the system resolves to a minimum of $\mathcal{F}$. Multiple minima may be equally accessible.
- **Crystallisation with observer**: the system resolves to a minimum of $\mathcal{F}$ that also satisfies $\mathcal{C}_\mathcal{O} \leq \epsilon$. The observer's coherence constraint biases the resolution toward configurations compatible with its persistence.

7.2 Observer participation, not determination

Remark 9. *The observer participates in crystallisation but does not uniquely determine the outcome. The observer constrains which configurations are admissible; the closure functional $\mathcal{F}$ determines which admissible configuration is selected. The observer is a boundary condition, not a decision-maker.*

7.3 Regime connection

Paper XXVIII [Smart, 2026d] proposed that the quantum and gravitational regimes correspond to weak-ordering and strong-ordering limits of the event set. The observer-constraint framework bridges these regimes:

- In the **weak-ordering (quantum) regime**, the observer constrains which of the multiple admissible configurations are accessible. This is measurement.
- In the **strong-ordering (gravitational) regime**, the observer's embedding within the ordered event set defines a frame. This is relativistic observation.

Both are instances of the same structural mechanism: the observer, as a coherent substructure, imposes constraints on the configurations accessible from its location within the event-order geometry.

8 Implications for the Locus of Experience

This section is brief and controlled. It addresses one narrow question: does the observer-constraint framework provide a *locus* for conscious experience within the dynamics?

8.1 What the framework provides

The observer, as defined in Section 3, has structural properties that are commonly associated with conscious systems:

- **Boundedness**: the observer is a finite, localised substructure, not a global property.
- **Internal coherence**: the observer maintains self-consistent internal state (stability condition).
- **Self-reference**: the observer's constraint functional depends on its own internal configuration.
- **Persistence**: the observer endures across the event ordering.
- **Sensitivity**: the observer is affected by its environment through boundary conditions.

The framework identifies these as necessary structural conditions for locatable experience: a substructure that is bounded, persistent, internally coherent, self-referential, and environmentally sensitive has the right structural features to serve as a natural locus of experience within the dynamics. These conditions are necessary, not sufficient: they do not by themselves explain qualia or subjective unity.

8.2 What the framework does not provide

- No explanation of qualia (the qualitative character of experience).
- No derivation of subjective unity or the binding problem.
- No claim that all observer regions are conscious, or that consciousness requires observer structure.
- No measurement of consciousness or threshold criterion.

8.3 The placement claim

> The framework identifies a class of necessary structural conditions for locatable experience: boundedness, persistence, internal coherence, self-referential organisation, and boundary-coupled environmental sensitivity. Substructures satisfying these conditions are natural candidates for the locus of experience within the dynamics. The framework does not explain experience. It makes experience *locatable* within the physics, rather than external to it.

Remark 10 (Epistemic status)**.** *This section identifies a structural correspondence, not a derivation. The claim is that the observer-constraint framework provides a natural locus for experience, not that it explains experience. The distinction between "locatable" and "explained" is the boundary of this paper.*

9 Minimal Toy Model

The following example illustrates the constraint-induced outcome restriction in the simplest possible setting. This toy model is illustrative only: it is not claimed to arise directly from the 600-cell construction or from the full event-order dynamics.

Example 11 (Two-configuration system with observer constraint)**.** *Consider an event set $\mathcal{E}$ with three events $\{e_1, e_2, e_3\}$ forming a path graph $e_1 - e_2 - e_3$. Suppose the closure functional $\mathcal{F}$ has two locally stable configurations: Φ_A and Φ_B, both satisfying $\delta\mathcal{F} = 0$, $\delta^2\mathcal{F} > 0$.*

An observer region $\mathcal{O} = \{e_2\}$ (the central event) carries a constraint functional:

$$\mathcal{C}_{\mathcal{O}}(\Phi|_{\mathcal{O}}, \partial_{\mathcal{O}}\Phi) = \left|\Phi(e_2) - \tfrac{1}{2}[\Phi(e_1) + \Phi(e_3)]\right|^2 \tag{5}$$

This measures the deviation of the observer's field value from the mean of its neighbours: the observer is coherent when it is locally consistent with its boundary.

Setup. *Configuration Φ_A has $\Phi_A(e_1) = 1$, $\Phi_A(e_2) = 1$, $\Phi_A(e_3) = 1$ (uniform). Configuration Φ_B has $\Phi_B(e_1) = 1$, $\Phi_B(e_2) = -1$, $\Phi_B(e_3) = 1$ (sign flip at observer).*

Constraint evaluation.

- $\mathcal{C}_{\mathcal{O}}\big(\Phi_A|_{\mathcal{O}}, \partial_{\mathcal{O}}\Phi_A\big) = |1 - \frac{1}{2}(1+1)|^2 = 0 \leq \epsilon$. *Admissible.*
- $\mathcal{C}_{\mathcal{O}}\big(\Phi_B|_{\mathcal{O}}, \partial_{\mathcal{O}}\Phi_B\big) = |-1 - \frac{1}{2}(1+1)|^2 = 4 > \epsilon$ *(for any reasonable $\epsilon < 4$). Not admissible.*

Result. *Without the observer, both Φ_A and Φ_B are stable configurations. With the observer constraint, only Φ_A is admissible. The observer's coherence requirement excludes the locally inconsistent configuration.*

This is the minimal structural analogue of measurement-induced outcome stabilisation: the observer does not choose Φ_A; its existence as a coherent substructure makes Φ_B dynamically inaccessible. The example does not establish that all physical measurement reduces to this form; it demonstrates only the minimal structural mechanism by which observer coherence can reduce admissible multiplicity.

10 Discussion and Limitations

10.1 What is established

- The observer is explicitly defined as a bounded, stable, self-referential substructure within the event-order geometry (Definition 2).
- A constraint functional $\mathcal{C}_{\mathcal{O}}$ formalises internal coherence.
- Observer-admissible configurations are defined (Definition 4).
- Quantum measurement is mapped to constraint-induced reduction of configuration multiplicity.
- Relativistic frame dependence is mapped to observer embedding within the event partial order.
- Both roles are unified as aspects of a single constraint-placement mechanism.

- Gödel's theorems are correctly scoped to formal symbolic systems.
- Crystallisation and observer constraint are distinguished (global vs. local).
- Necessary structural conditions for locatable experience are identified (not sufficient for explaining qualia).
- A minimal toy model demonstrates constraint-induced outcome restriction (Example 11).

10.2 What is not established

- No derivation of the Born rule or specific outcome probabilities.
- No construction of a concrete $\mathcal{C}_{\mathcal{O}}$ from the 600-cell geometry (the constraint functional is defined abstractly).
- No proof that the observer constraint is unique or necessary for state selection.
- No theory of consciousness, qualia, or subjective binding.
- No experimental predictions from the observer-placement framework itself.
- No proof that observer regions satisfying Definition 2 exist in physically realistic configurations (existence is assumed, not derived).
- The coherence threshold ϵ is not determined by the framework.

10.3 Relation to other approaches

Table 3: Observer placement across interpretive frameworks.

Framework	Observer status	Placement
Copenhagen QM	External, undefined	Outside the formalism
Many-worlds	Emergent branch-relative subsystem	Branch-relative, not singled out
Relational QM	Relative to other systems	No internal structure
GR (standard kinematics)	Idealised worldline	No internal structure
Integrated Information	Φ-maximising system	Information-theoretic
VFD (this paper)	Constraint substructure	Inside the dynamics

11 Conclusion

The observer problem in physics is not the absence of explanation but the absence of placement. Quantum mechanics registers definite outcomes relative to measurement interactions without supplying a universally agreed internal placement of the observer. General relativity defines physics relative to a frame without giving the frame-defining entity internal structure. The explanatory gap in consciousness studies shares the same structural feature: the observer is assumed to exist but not located within the dynamics.

This paper proposes a placement. The observer is a bounded, dynamically stable, self-referential substructure within the event-order geometry of the VFD framework. Its internal coherence, formalised through the constraint functional $\mathcal{C}_{\mathcal{O}}$, imposes conditions on which field configurations are admissible. In the quantum regime, this constraint constrains superposed multiplicity toward definite outcomes. In the gravitational regime, the observer's embedding within the event partial order defines a relativistic frame. Both roles are aspects of a single mechanism: the observer, as a coherent substructure, constrains the accessible configurations from its position within the geometry.

The framework makes conscious experience locatable — associated with bounded, persistent, self-referential constraint structures — without claiming to explain it.

What was missing was not the physics but the placement.

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